RTL Design Sherpa

APB RTC Micro-Architecture Specification 1.0

January 4, 2026

Table of Contents

List of Figures

List of Tables

No tables in this document.

List of Waveforms

1 Rtc Mas Index

Generated: 2026-08-14

RTL Design Sherpa · Learning Hardware Design Through Practice GitHub · Documentation Index · MIT License

---

2 APB RTC - Overview

2.1 Introduction

The APB RTC is a Real-Time Clock controller with an APB slave interface. It maintains time and date and provides alarm and 1 Hz tick interrupt capabilities.

2.2 Key Features

2.2.1 Time Keeping

- Seconds, minutes, hours (12/24-hour mode)
- Day of month (1-31), month, year (0-99, base year 2000 is hardcoded)
- Leap year calculation (base 2000, valid through 2099); the year field wraps 99 to 00 with no century carry
- Binary format by default, optional BCD format

2.2.2 Alarm Function

- Single programmable alarm
- Seconds, minutes, hours match with per-field match enables

2.2.3 Interrupt Support

- Alarm match interrupt
- Second-tick interrupt (fixed 1 Hz)

2.2.4 Power Management

- Low-power 32.768 kHz oscillator

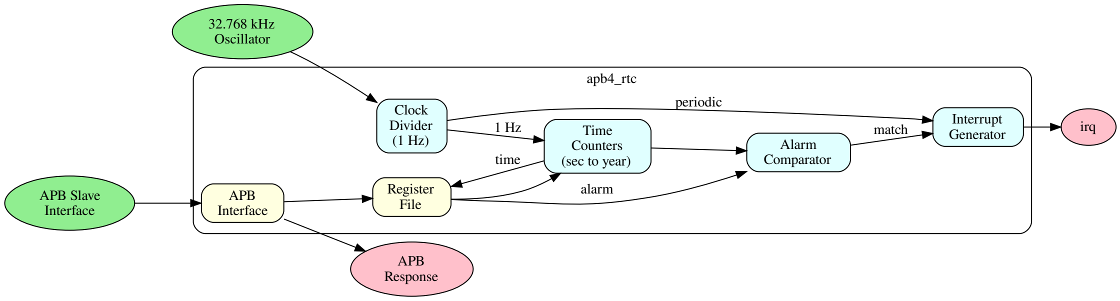
2.3 Applications

- System timekeeping

- Scheduled wake-up
- Event timestamping
- Calendar functions
- Alarm clock

2.4 Block Diagram

2.4.1 Figure 1.1: RTC Block Diagram

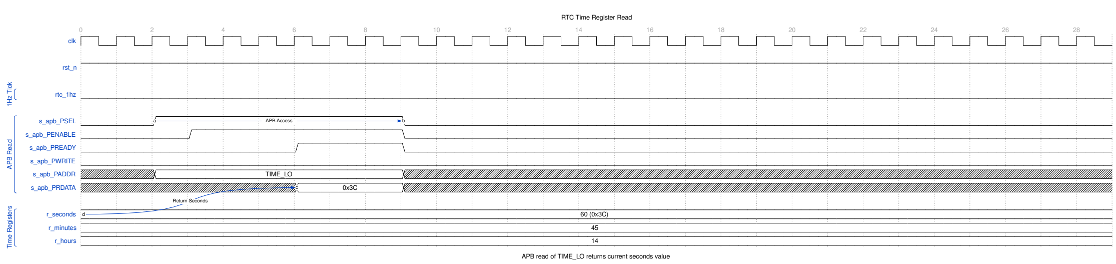


RTC Block Diagram

2.5 Timing Diagrams

2.5.1 Waveform 1.1: Time Register Read

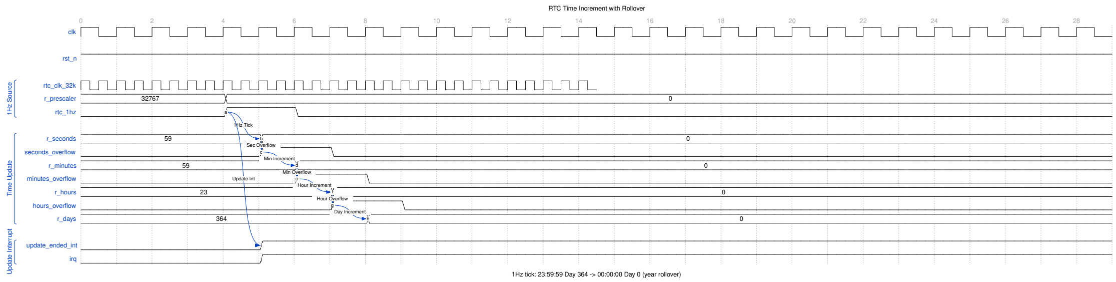
Reading the time registers returns the current time value.



RTC Time Read

2.5.2 Waveform 1.2: Time Increment with Rollover

Shows the cascade of time registers as seconds overflow to minutes, minutes to hours, etc.

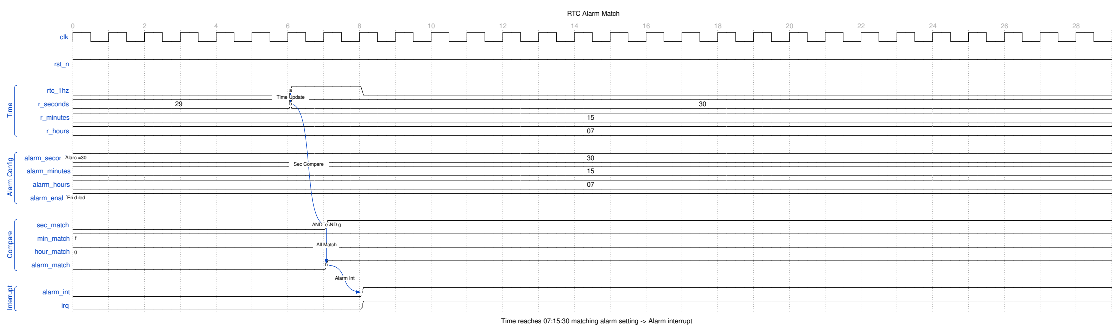


RTC Time Increment

The 1Hz tick from the 32.768kHz prescaler triggers the seconds counter. Each overflow cascades to the next register, demonstrating the 23:59:59 to 00:00:00 rollover.

2.5.3 Waveform 1.3: Alarm Match

When the current time matches the alarm setting, an interrupt is generated.

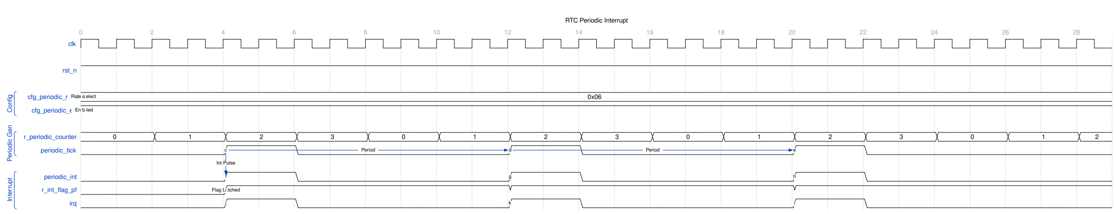


RTC Alarm Match

All configured alarm fields (seconds, minutes, hours) must match simultaneously for the alarm to trigger.

2.5.4 Waveform 1.4: Second-Tick Interrupt

The RTC generates a fixed 1 Hz tick interrupt when enabled by `second_int_enable`.



RTC Periodic Interrupt

The 1 Hz tick is derived from the 32.768 kHz oscillator by a fixed divide-by-32768; there is no programmable rate selector. Each tick sets the `second_tick` status flag and, when enabled, asserts the second-tick interrupt.

2.6 Register Summary

Offset	Name	Access	Description
0x00	RTC_CONFIG	RW	Global configuration (enable, hour/BCD/clock mode, time-set)
0x04	RTC_CONTROL	RW	Alarm and interrupt enables
0x08	RTC_STATUS	RO/W1C	Status flags and indicators
0x0C	RTC_SECONDS	RW	Seconds (0-59)
0x10	RTC_MINUTES	RW	Minutes (0-59)
0x14	RTC_HOURS	RW	Hours (0-23 or 1-12)
0x18	RTC_DAY	RW	Day of month (1-31)
0x1C	RTC_MONTH	RW	Month (1-12)
0x20	RTC_YEAR	RW	Year (0-99, base 2000)
0x24	RTC_ALARM_SEC	RW	Alarm seconds
0x28	RTC_ALARM_MIN	RW	Alarm minutes
0x2C	RTC_ALARM_HOUR	RW	Alarm hours
0x30	RTC_ALARM_MASK	RW	Alarm field match enables

See ch05 Register Map for full bit-level definitions and the time-set protocol.

Next: 02_architecture.md - Architecture details

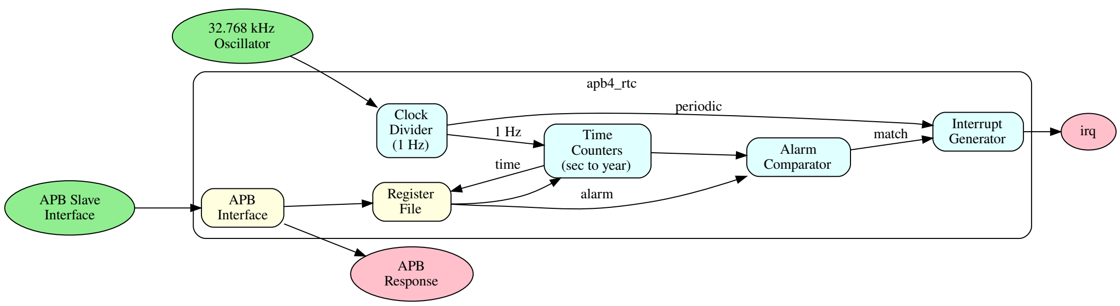
RTL Design Sherpa · Learning Hardware Design Through Practice GitHub · Documentation Index · MIT License

---

3 APB RTC - Architecture

3.1 High-Level Block Diagram

3.1.1 Figure 1.2: RTC Architecture

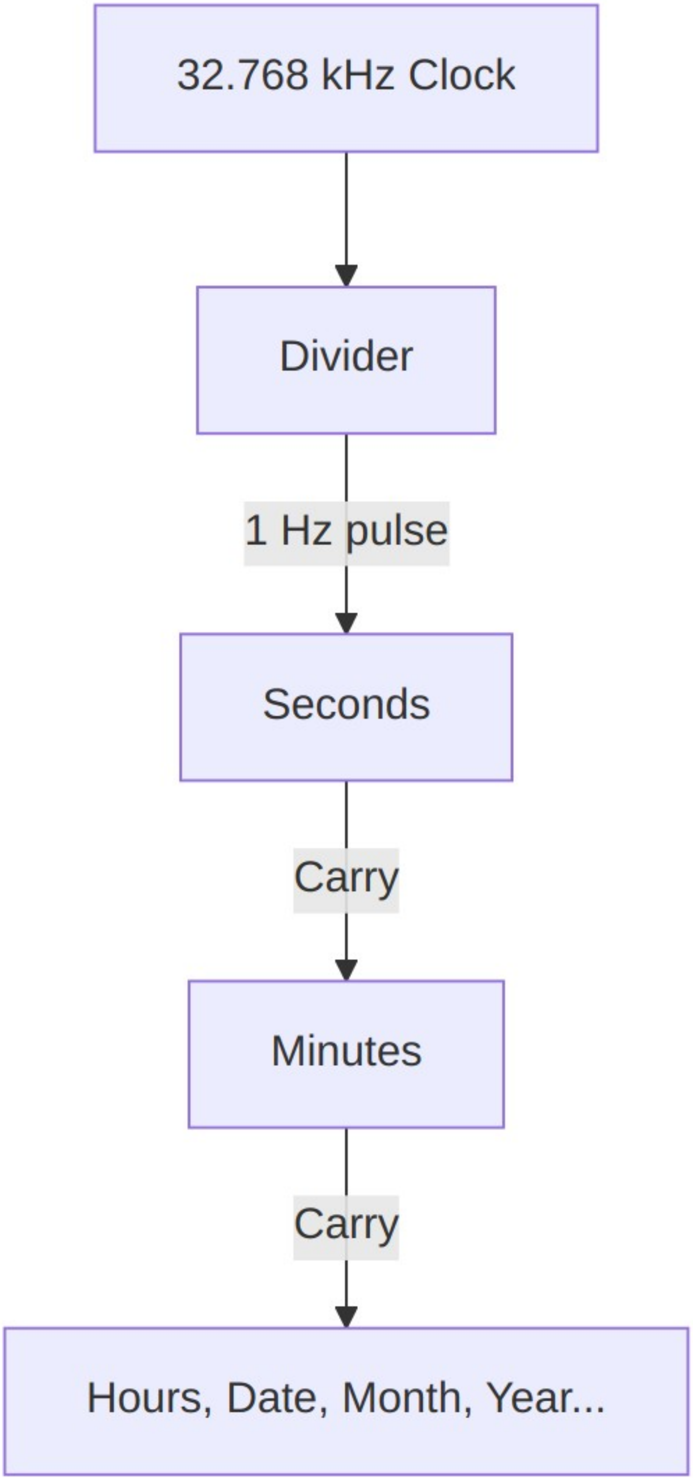


RTC Architecture

3.2 Module Hierarchy

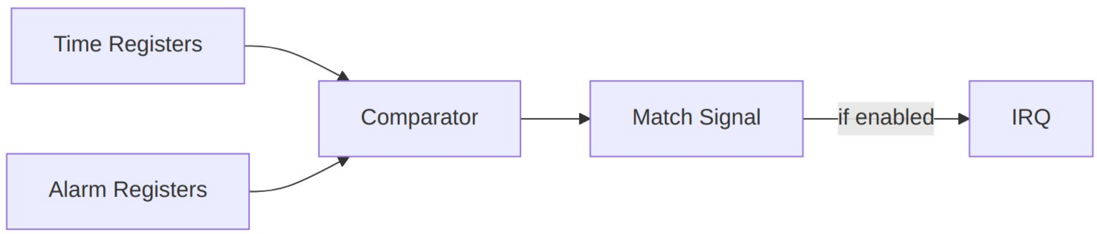
```
apb4_rtc (Top Level)
+-- apb4_slave
+-- rtc_config_regs (Register Wrapper)
|   +-- rtc_regs (PeakRDL Generated)
|
+-- rtc_core
    +-- Time Counter (seconds to year)
    +-- Alarm Comparator
    +-- Interrupt Generator
    +-- BCD Logic
```

3.3 Data Flow



32.768 kHz Clock
Divider
1 Hz pulse
Seconds
Carry
Minutes
Carry
Hours, Date, Month, Year...

Time Update Flow



Alarm Match Flow

3.4 Clock Domains

- APB domain (pclk): Register access
- RTC domain (32.768 kHz): Time counting
- CDC when clocks are asynchronous

Next: 03_clocks_and_reset.md

---

4 APB RTC - Register Map

The register block occupies 0x34 bytes (13 registers). Only address bits [5:0] are decoded, so any read at 0x34 or above returns 0 with no error.

4.1 Register Summary

Offset	Name	Access	Reset	Description
0x00	RTC_CONFIG	RW	0x00000000	Global configuration (enable, hour/BCD/clock mode, time-set)
0x04	RTC_CONTROL	RW	0x00000000	Alarm and interrupt enables
0x08	RTC_STA	RO/W1C	0x0000000	Status flags and

Offset	Name	Access	Reset	Description
	TUS		0	indicators
0x0C	RTC_SECONDS	RW	0x00	Seconds (0-59)
0x10	RTC_MINUTES	RW	0x00	Minutes (0-59)
0x14	RTC_HOURS	RW	0x00	Hours (0-23, or 1-12 + PM in 12-hour mode)
0x18	RTC_DAY	RW	0x01	Day of month (1-31)
0x1C	RTC_MONTH	RW	0x01	Month (1-12)
0x20	RTC_YEAR	RW	0x00	Year (0-99, base 2000)
0x24	RTC_ALARM_SEC	RW	0x00	Alarm seconds match value
0x28	RTC_ALARM_MIN	RW	0x00	Alarm minutes match value
0x2C	RTC_ALARM_HOUR	RW	0x00	Alarm hours match value
0x30	RTC_ALARM_MASK	RW	0x00	Alarm field match enables

All registers are 32 bits wide. In each register only the low bits listed below are implemented; the remaining bits are reserved (read as 0).

4.2 RTC_CONFIG (0x00)

Bit	Name	Access	Reset	Description
0	rtc_enable	RW	0	Master enable for the RTC (0=disabled, 1=enabled)
1	hour_mode_12	RW	0	Hour format: 0=24-hour, 1=12-hour (AM/PM)

Bit	Name	Access	Reset	Description
2	bcd_mode	RW	0	Time format: 0=binary, 1=BCD
3	clock_select	RW	0	Time-counter clock: 0=32.768 kHz, 1=system clock (test)
4	time_set_mode	RW	0	1=allow time setting (stops the counter), 0=normal operation
31:5	Reserved	RO	0	Reserved

Note that `hour_mode_12` and the format/clock controls live here, not in RTC_CONTROL. Out of reset the counters run in **binary** mode; BCD is optional and is selected by setting `bcd_mode`.

4.2.1 Setting the time (time_set_mode protocol)

Time registers are mirrors of the internal counters: a plain write to RTC_SECONDS..RTC_YEAR is reloaded from the counter on the next cycle unless the time-set protocol is used. To set the time:

1. Write RTC_CONFIG with `rtc_enable = 1` and `time_set_mode = 1` (both are required; the load path is gated on the RTC being enabled). Also select the desired `hour_mode_12` and `bcd_mode`. While `time_set_mode = 1` the counter is stopped.
2. Write the new values to RTC_SECONDS, RTC_MINUTES, RTC_HOURS, RTC_DAY, RTC_MONTH, and RTC_YEAR. A write to any time register (0x0C-0x20) latches the full set into the counters and sets `time_valid`.
3. Write RTC_CONFIG again with `time_set_mode = 0` to resume counting.

4.3 RTC_CONTROL (0x04)

Bit	Name	Access	Reset	Description
0	alarm_enable	RW	0	Enable alarm comparison
1	alarm_int_enable	RW	0	Enable interrupt on alarm match
2	second_int_enable	RW	0	Enable interrupt every second (1 Hz tick)
31:3	Reserved	RO	0	Reserved

4.4 RTC_STATUS (0x08)

Bit	Name	Access	Reset	Description
0	alarm_flag	W1C	0	Alarm triggered; write 1 to clear
1	second_tick	W1C	0	1 Hz tick occurred; write 1 to clear
2	time_valid	RO	0	Time registers contain valid data (set after time-set)
3	pm_indicator	RO	0	12-hour mode: 0=AM, 1=PM (BCD 12-hour mode only)
31:4	Reserved	RO	0	Reserved

4.5 RTC_ALARM_MASK (0x30)

Bit	Name	Access	Reset	Description
0	sec_match_en	RW	0	1=compare seconds for alarm, 0=don't care
1	min_match_en	RW	0	1=compare minutes for alarm, 0=don't care
2	hour_match_en	RW	0	1=compare hours for alarm, 0=don't care
31:3	Reserved	RO	0	Reserved

The mask bits are active-high *match enables*, not "ignore" bits. Out of reset all three are 0, so every field is a don't-care and an enabled alarm matches every second. Set the relevant `*_match_en` bits for the fields that must match.

4.6 Time Format

Time/date values are stored in binary by default. When `bcd_mode` (RTC_CONFIG bit 2) is set, the same fields are stored in BCD:

Field	Binary	BCD
Seconds	0-59	0x00-0x59
Minutes	0-59	0x00-0x59
Hours (24-hour)	0-23	0x00-0x23
Hours (12-hour)	1-12	0x01-0x12
Day of month	1-31	0x01-0x31
Month	1-12	0x01-0x12
Year	0-99	0x00-0x99

In 12-hour BCD mode, RTC_HOURS bit 7 carries the PM indicator.

Back to: RTC Specification Index

---

5 Retro Legacy Blocks - Product Requirements Document

Component: Retro Legacy Blocks (RLB) - Production-Quality Legacy Peripherals **Version:** 1.0 **Status:** 🟢 Active Development - HPET Production Ready **Last Updated:** 2025-10-29

5.1 1. Overview

5.1.1 1.1 Purpose

The Retro Legacy Blocks (RLB) component provides production-quality implementations of legacy peripheral blocks based on proven peripheral designs. These blocks are designed to be reusable, well-tested, and suitable for both FPGA and ASIC implementation.

5.1.2 1.2 Design Philosophy

"Retro" - Proven Architectures: - Implements time-tested peripheral designs from successful platforms - Focuses on simplicity, reliability, and well-understood behavior - Prioritizes production-readiness over experimental features

"Legacy" - Time-Tested Interfaces: - Based on proven peripheral interface specifications - Suitable for systems requiring retro-compatible peripheral compatibility - APB-based interface for easy integration

"Blocks" - Modular Collection: - Each peripheral is independent and self-contained - Clear separation between different blocks (rtl/hpet/, rtl/gpio/, etc.) - Can be used individually or wrapped into integrated subsystem

5.1.3 1.3 Target Applications

- Retro-compatible platform compatibility layers
- Embedded systems requiring legacy peripheral interfaces
- FPGA-based system emulation
- Educational platforms demonstrating classic peripheral designs
- Mixed-vintage SoC integration (modern + legacy interfaces)

5.2 2. Implemented Blocks

5.2.1 2.1 HPET - High Precision Event Timer

Status: √ Production Ready (5/6 configurations 100% passing) **RTL Location:** `rtl/hpet/` **Documentation:** `docs/hpet_mas/`

Key Features: - Configurable timer count: 2, 3, or 8 independent timers - 64-bit main counter for high-resolution timestamps - 64-bit comparators per timer - Operating modes: One-shot and periodic - Clock domain crossing: Optional CDC for timer/APB clock independence - APB4 interface: Standard AMBA APB protocol - PeakRDL integration: Register map generated from SystemRDL specification

Applications: - System tick generation - Real-time OS scheduling - Precise event timing - Performance profiling - Watchdog timers - Multi-rate timing domains

Test Coverage: - 6 configurations tested (2/3/8 timers, CDC on/off) - 5/6 configurations at 100% pass rate - 1 configuration at 92% (minor stress test timeout) - 12 test cases per configuration (basic/medium/full)

Architecture:

```
┌─────────────────────────────────────────────────────────────┐
│                         APB HPET                             │
```

```
┌──────────────────────────────────────────────────────┐
│                                                      │
│  ┌──────────────┐    ┌──────────────┐    ┌──────────────┐  │
│  │ APB Slave    │───▶│ Config Regs  │───▶│  HPET Core   │  │
│  │ (Optional    │    │ (PeakRDL)    │    │              │  │
│  │  CDC)        │    │              │    │ - Counter    │  │
│  └──────────────┘    └──────────────┘    │ - Timers[N]  │  │
│                                          │  - Comparators│ │
│                                          └──────────────┘  │
│                                                 │          │
│                                       Timer IRQs [N:0]     │
└──────────────────────────────────────────────────────┘
```

Design Highlights: - Reset macro standardization (FPGA-friendly) - Per-timer data buses prevent corruption - Edge-triggered register write strobes (not level) - W1C status register for interrupt clearing - Optional asynchronous clock domains with handshake CDC

See: `docs/hpet_mas/hpet_mas_index.md` for complete HPET specification

5.3 3. Planned Blocks

5.3.1 3.1 8259 - Programmable Interrupt Controller (PIC)

Status: 📋 Planned **Priority:** High **Effort:** 6-8 weeks **Address:** `0x4000_1000 - 0x4000_1FFF` (4KB window)

Planned Features: - Intel 8259A-compatible register interface - 8 interrupt request (IRQ) inputs - Cascadable (master/slave configuration) - Priority resolver (fixed and rotating priority) - Edge and level triggered modes - Interrupt mask register - End-of-Interrupt (EOI) handling - APB register interface

Applications: - Legacy interrupt management - PC-compatible systems - Hardware interrupt aggregation - Priority-based interrupt handling - Cascaded multi-level interrupt systems

5.3.2 3.2 8254 - Programmable Interval Timer (PIT)

Status: 📋 Planned **Priority:** High **Effort:** 4-5 weeks **Address:** `0x4000_2000 - 0x4000_2FFF` (4KB window)

Planned Features: - Intel 8254-compatible register interface - 3 independent 16-bit counters - 6 programmable counter modes - Binary and BCD counting - Read-back command - Configurable clock input - Interrupt/output generation per counter - APB register interface

Counter Modes: - Mode 0: Interrupt on terminal count - Mode 1: Hardware retriggerable one-shot - Mode 2: Rate generator - Mode 3: Square wave mode - Mode 4: Software triggered strobe - Mode 5: Hardware triggered strobe

Applications: - System tick generation - Periodic timer interrupts - Square wave generation - Event counting - Legacy PC timer compatibility

5.3.3 3.3 GPIO - General Purpose I/O

Status: 📋 Planned **Priority:** Medium **Effort:** 4-6 weeks **Address:** TBD (not in primary ILB address map)

Planned Features: - Configurable pin count (8, 16, 32 pins) - Per-pin direction control (input/output/bidirectional) - Input debouncing logic - Interrupt generation (rising/falling/both edges, level) - Output drive strength configuration - Pull-up/pull-down control - APB register interface

Applications: - LED control - Button inputs - Hardware control signals - Chip-select generation - Status monitoring

5.3.4 3.4 RTC - Real-Time Clock

Status: 📋 Planned **Priority:** Medium **Effort:** 3-4 weeks **Address:** `0x4000_3000 - 0x4000_3FFF` (4KB window)

Planned Features: - 32.768 kHz clock input (typical RTC crystal frequency) - Seconds, minutes, hours, day, month, year tracking - Alarm functionality - Battery backup support (power domain considerations) - 24-hour or 12-hour (AM/PM) mode - Leap year handling - APB register interface

Applications: - System time-of-day tracking - Wake-on-alarm functionality - Timestamp generation - Power-aware applications

5.3.5 3.5 SMBus Controller

Status: 📋 Planned **Priority:** Medium **Effort:** 6-8 weeks **Address:** `0x4000_4000 - 0x4000_4FFF` (4KB window)

Planned Features: - SMBus 2.0 compliance - Master and slave modes - Clock stretching support - Packet Error Checking (PEC) - Alert response address - Configurable clock speed - APB register interface

Applications: - System management bus communication - Sensor interfaces (temperature, voltage) - EEPROM access - Battery management - Fan control

5.3.6 3.6 UART - Universal Asynchronous Receiver/Transmitter

Status: 📋 Planned **Priority:** Medium **Effort:** 4-5 weeks **Address:** TBD (not in primary ILB address map)

Planned Features: - 16550-compatible register interface - Configurable baud rate generation - 5/6/7/8 data bits - Parity: none, even, odd, mark, space - Stop bits: 1, 1.5, 2 - Hardware flow control (RTS/CTS) - FIFO buffers (16-byte TX/RX) - Interrupt generation

Applications: - Debug console - Serial communication - Modem interfaces - Legacy peripheral communication

5.3.7 3.7 SPI Controller

Status: 📋 Planned **Priority:** Low **Effort:** 5-6 weeks **Address:** TBD (not in primary ILB address map)

Planned Features: - Master mode (initially; slave mode future) - Configurable clock polarity and phase (CPOL/CPHA) - Multiple chip selects - Configurable word size (8/16/32 bits) - TX/RX FIFOs - DMA support (future) - APB register interface

Applications: - Flash memory access - ADC/DAC interfaces - Display controllers - SD card communication

5.3.8 3.8 I2C Controller

Status: 📋 Planned **Priority:** Low **Effort:** 5-7 weeks **Address:** TBD (not in primary ILB address map)

Planned Features: - I2C standard (100 kHz), fast (400 kHz), fast-plus (1 MHz) modes - Multi-master arbitration - 7-bit and 10-bit addressing - Clock stretching - General call support - APB register interface

Applications: - Sensor interfaces - EEPROM access - Multi-chip communication - System configuration

5.3.9 3.9 Watchdog Timer

Status: 📋 Planned **Priority:** Low **Effort:** 2-3 weeks **Address:** TBD (not in primary ILB address map)

Planned Features: - Configurable timeout period - Countdown counter with reload - Reset generation on timeout - Lock mechanism to prevent accidental disable - Interrupt before reset (optional warning) - APB register interface

Applications: - System fault recovery - Software hang detection - Periodic system reset - Safety-critical applications

5.3.10 3.10 Power Management / ACPI Controller

Status: 📋 Planned **Priority:** Medium **Effort:** 8-10 weeks **Address:** `0x4000_5000 - 0x4000_5FFF` (4KB window)

Planned Features: - Clock gating control per block - Power domain sequencing - Reset generation and distribution - Wake event handling - Sleep/idle mode control - ACPI-compatible registers - APB register interface

Applications: - Low-power system design - Battery-powered devices - Dynamic power management - Thermal management - OS power management interface

5.3.11 3.11 IOAPIC - I/O Advanced Programmable Interrupt Controller

Status: 📋 Planned **Priority:** Medium **Effort:** 6-8 weeks **Address:** `0x4000_6000 - 0x4000_6FFF` (4KB window)

Planned Features: - I/O APIC CSR model (register-based interface) - Multiple interrupt inputs (24+) - Programmable interrupt routing - Edge and level triggered modes - Priority-based arbitration - Interrupt masking per input - APB register interface for configuration

Applications: - Advanced interrupt routing - Multi-processor interrupt distribution - Flexible interrupt mapping - Legacy IRQ redirection - PC-compatible systems

5.3.12 3.12 Interconnect ID / Version Registers

Status: 📋 Planned **Priority:** Low **Effort:** 1-2 weeks **Address:** `0x4000_F000 - 0x4000_FFFF` (4KB window)

Planned Features: - Vendor ID register - Device ID register - Revision ID register - Block presence/capability bits - Configuration status registers - Debug/diagnostic registers - APB register interface

Applications: - Software block discovery - Version checking - Feature detection - Debug and diagnostics - Platform identification

---

5.4 4. Integration and Wrapper Goals

5.4.1 4.1 Individual Block Integration

Each block is designed to be used standalone:

Example - HPET Integration:

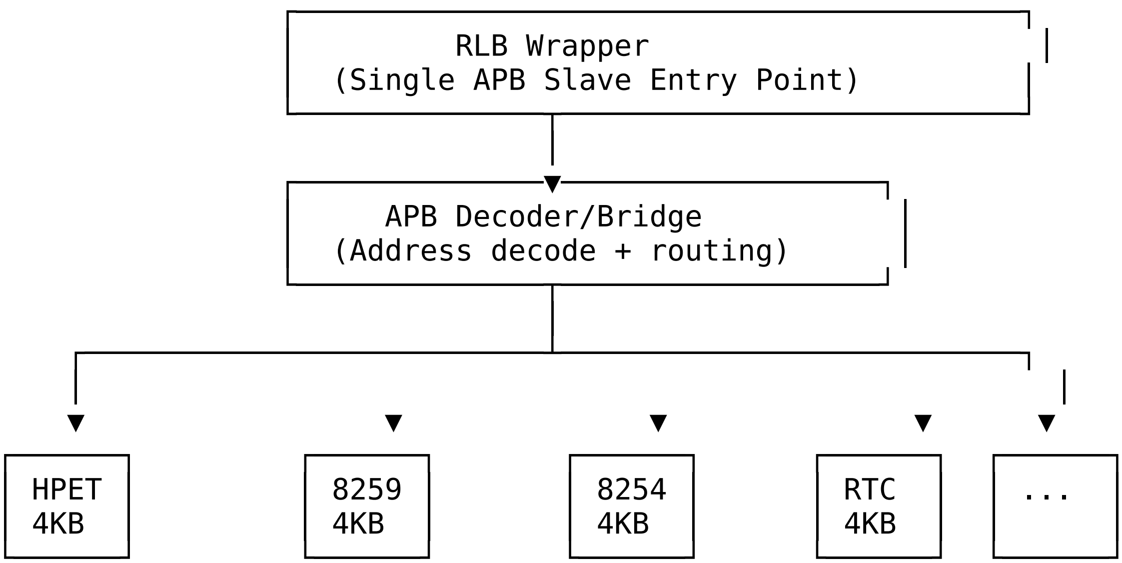
```
apb4_hpet #(
    .NUM_TIMERS(3),
    .VENDOR_ID(16'h8086),
    .REVISION_ID(16'h0001),
    .CDC_ENABLE(0)
) u_hpet (
    .pclk          (apb_clk),
    .presetn       (apb_rst_n),
    // APB interface
    .paddr         (paddr),
    .psel          (psel_hpet),
    .penable       (penable),
    .pwrite        (pwrite),
    .pwdata        (pwdata),
    .prdata        (prdata_hpet),
```

```
    .pready       (pready_hpet),
    .pslverr      (pslverr_hpet),
    // HPET-specific
    .hpet_clk     (timer_clk),
    .hpet_rst_n   (timer_rst_n),
    .timer_irq    (timer_irq[2:0])
);
```

5.4.2 4.2 RLB Wrapper Architecture

Goal: Create top-level wrapper combining multiple legacy blocks into unified retro-compatible subsystem.

System Architecture:



Address Map:

Base address: `0x4000_0000` (1GB region in typical 32-bit system) Window size: 4KB per block (clean power-of-2 decode)

Address Range	Block	Size	Function
`0x4000_0000 - 0x4000_0FFF`	HPET	4KB	High Precision Event Timer
`0x4000_1000 - 0x4000_1FFF`	8259	4KB	Programmable Interrupt Controller (PIC)
`0x4000_2000 - 0x4000_2FFF`	8254	4KB	Programmable Interval Timer

Address Range	Block	Size	Function
			(PIT)
0x4000_3000 - 0x4000_3FFF	RTC	4KB	Real-Time Clock
0x4000_4000 - 0x4000_4FFF	SMBus	4KB	SMBus Host Controller
0x4000_5000 - 0x4000_5FFF	PM/ACPI	4KB	Power Management / ACPI Registers
0x4000_6000 - 0x4000_6FFF	IOAPIC	4KB	I/O Advanced PIC (CSR model)
0x4000_7000 - 0x4000_EFFF	*Reserved*	32KB	Future expansion
0x4000_F000 - 0x4000_FFFF	Interconnect	4KB	ID/Version/Control registers
All other addresses	Error Slave	-	Returns DECERR/SLVERR

Decoder Implementation:

```
// Address decode logic (simplified)
localparam BASE_ADDR = 32'h4000_0000;
localparam BLOCK_SIZE = 12;  // 4KB = 2^12

logic [3:0] block_sel;
assign block_sel = paddr[15:12];  // Extract window number

always_comb begin
    psel_hpet      = (block_sel == 4'h0) & psel;  // 0x4000_0xxx
    psel_pic8259   = (block_sel == 4'h1) & psel;  // 0x4000_1xxx
    psel_pit8254   = (block_sel == 4'h2) & psel;  // 0x4000_2xxx
    psel_rtc       = (block_sel == 4'h3) & psel;  // 0x4000_3xxx
    psel_smbus     = (block_sel == 4'h4) & psel;  // 0x4000_4xxx
    psel_pm        = (block_sel == 4'h5) & psel;  // 0x4000_5xxx
    psel_ioapic    = (block_sel == 4'h6) & psel;  // 0x4000_6xxx
    psel_id        = (block_sel == 4'hF) & psel;  // 0x4000_Fxxx
    psel_error     = !(|{psel_hpet, psel_pic8259, psel_pit8254,
                         psel_rtc, psel_smbus, psel_pm,
                         psel_ioapic, psel_id}) & psel;
end
```

Interface: - **Single APB slave port** at base address 0x4000_0000 - **Aggregated interrupt output** combining all block IRQs - **Per-block clock/reset control** for power management - **External I/O signals** (GPIO, UART, I2C/SMBus, etc.) - **Error slave** returns SLVERR for unmapped addresses

Benefits: - Simplified system integration (single APB slave) - Consistent 4KB window addressing - Clean power-of-2 address decode - Easy expansion (32KB reserved space) - Single verification target - Drop-in retro-compatible peripheral subsystem

---

5.5 5. Design Standards

5.5.1 5.1 Reset Handling

MANDATORY: All blocks must use standardized reset macros from `rtl/amba/includes/reset_defs.svh`

Pattern:

```
`include "reset_defs.svh"

`ALWAYS_FF_RST(clk, rst_n,
    if (`RST_ASSERTED(rst_n)) begin
        r_state <= IDLE;
        r_counter <= '0;
    end else begin
        r_state <= w_next_state;
        r_counter <= r_counter + 1'b1;
    end
)
```

Why: - FPGA-friendly reset inference - Consistent synthesis behavior - Single-point reset polarity control - Better timing closure

5.5.2 5.2 Register Generation

Preferred: Use PeakRDL for register map generation

Process: 1. Define registers in SystemRDL (`.rdl` file) 2. Generate RTL using PeakRDL regblock 3. Create wrapper module connecting registers to core logic 4. Use edge detection for write strobes (not level)

Benefits: - Consistent register interface - Auto-generated documentation - Reduced manual RTL errors - Easy register map changes

5.5.3 5.3 Testbench Architecture

MANDATORY: Follow project testbench organization pattern

Structure:

```
dv/
├── tbclasses/{block}/          # Block-specific TB classes
```

```
│   ├── {block}_tb.py          # Main testbench
│   ├── {block}_tests_basic.py # Basic test suite
│   ├── {block}_tests_medium.py # Medium test suite
│   └── {block}_tests_full.py  # Full test suite
└── tests/{block}/             # Test runners
    ├── test_apb_{block}.py    # Pytest wrapper
    └── conftest.py            # Pytest configuration
```

Import Pattern:

```python
# Always import from PROJECT AREA
from projects.components.retro_legacy_blocks.dv.tbclasses.{block}.
{block}_tb import {Block}TB
```

Test Levels: - **Basic:** Core functionality (register access, basic operation) - **Medium:** Extended features (modes, configurations, edge cases) - **Full:** Stress testing, CDC variants, corner cases

Target: 100% pass rate at all levels

5.5.4 5.4 FPGA Synthesis Attributes

MANDATORY: Add FPGA synthesis hints for memory arrays

```systemverilog
`ifdef XILINX
    (* ram_style = "auto" *)
`elsif INTEL
    /* synthesis ramstyle = "AUTO" */
`endif
logic [DATA_WIDTH-1:0] mem [DEPTH];
```

5.5.5 5.5 Documentation Requirements

Each block must have: - RTL comments (inline) - Register map specification - Block-level specification in `docs/{block}_mas/` - Integration guide - Test plan and results

---

5.6 6. Quality Metrics

5.6.1 6.1 Production Readiness Criteria

A block is considered "Production Ready" when:

- √ All basic tests pass 100%
- √ All medium tests pass 100%
- √ All full tests pass ≥95%
- √ Complete register map specification
- √ RTL lint clean (Verilator)

- √ Reset macros used throughout
- √ FPGA synthesis attributes applied
- √ Integration guide written
- √ Known issues documented

5.6.2 6.2 Current Status

Block	Priority	Status	Test Pass Rate	Documentation	Production Ready
HPET	High	√ Complete	5/6 at 100%, 1/6 at 92%	√ Complete	√ Yes
8259 PIC	High	📋 Planned	N/A	N/A	✗ No
8254 PIT	High	📋 Planned	N/A	N/A	✗ No
GPIO	Medium	📋 Planned	N/A	N/A	✗ No
RTC	Medium	📋 Planned	N/A	N/A	✗ No
SMBus	Medium	📋 Planned	N/A	N/A	✗ No
PM/ACPI	Medium	📋 Planned	N/A	N/A	✗ No
IOAPIC	Medium	📋 Planned	N/A	N/A	✗ No
UART	Medium	📋 Planned	N/A	N/A	✗ No
SPI	Low	📋	N/A	N/A	✗ No

Block	Priority	Status	Test Pass Rate	Documentation	Production Ready
		Planned			
I2C	Low	📋 Planned	N/A	N/A	✗ No
Watchdog	Low	📋 Planned	N/A	N/A	✗ No
Interconnect	Low	📋 Planned	N/A	N/A	✗ No

5.7 7. Development Roadmap

5.7.1 7.1 Phase 1: Foundation (Complete ✓)

- √ HPET implementation
- √ Directory structure for multiple blocks
- √ Testbench architecture established
- √ Documentation templates
- √ Build and test infrastructure

5.7.2 7.2 Phase 2: Core Peripherals (Next 6-9 Months)

Q1 2026 (High Priority): - 8259 PIC (6-8 weeks) - Interrupt controller - 8254 PIT (4-5 weeks) - Interval timer - RTC (3-4 weeks) - Real-time clock

Q2 2026 (Medium Priority): - GPIO Controller (4-6 weeks) - SMBus Controller (6-8 weeks) - PM/ACPI Controller (8-10 weeks)

Q3 2026: - UART (4-5 weeks) - IOAPIC (6-8 weeks)

5.7.3 7.3 Phase 3: Advanced Peripherals (9-15 Months)

Q4 2026: - SPI Controller (5-6 weeks) - I2C Controller (5-7 weeks) - Watchdog Timer (2-3 weeks)

Q1 2027: - Interconnect ID/Version Registers (1-2 weeks) - ILB Wrapper integration starts

5.7.4 7.4 Phase 4: System Integration (15+ Months)

Q2-Q4 2027: - Complete ILB wrapper with all blocks - System-level integration examples - Performance characterization - FPGA reference designs - Application notes - Software driver examples

---

5.8 8. References

5.8.1 8.1 External Standards

Peripheral Specifications: - ACPI HPET Specification 1.0a - SMBus Specification Version 2.0 - 16550 UART Datasheet - I2C Specification (NXP) - SPI Protocol Specification

Bus Protocols: - AMBA APB Protocol Specification (ARM) - AMBA 3 APB Protocol v1.0

5.8.2 8.2 Internal Documentation

- `/CLAUDE.md` - Repository AI guide
- `/PRD.md` - Master repository requirements
- `projects/components/retro_legacy_blocks/CLAUDE.md` - Component AI guide
- `projects/components/retro_legacy_blocks/README.md` - Component overview
- `projects/components/retro_legacy_blocks/TASKS.md` - Task tracking

5.8.3 8.3 Block-Specific Documentation

HPET: - `docs/hpet_mas/hpet_mas_index.md` - HPET specification - `docs/IMPLEMENTATION_STATUS.md` - HPET test results - `known_issues/` - HPET issue tracking

---

5.9 9. Success Criteria

5.9.1 9.1 Individual Block Success

Each block must: - Pass all basic/medium tests at 100% - Pass full tests at ≥95% - Have complete register map specification - Include integration guide with examples - Be lint-clean (Verilator) - Use reset macros throughout - Include FPGA synthesis attributes

5.9.2 9.2 Collection Success

The retro_legacy_blocks component is successful when: - At least 6 blocks production-ready (HPET + 5 high/medium priority blocks) - All blocks follow consistent architecture (reset macros, PeakRDL, APB interface) - RLB wrapper integrates all blocks seamlessly with clean 4KB addressing - System-level integration example provided - Complete documentation for all blocks - FPGA reference design available - Address map covers all essential retro-compatible peripherals

5.9.3 9.3 Long-Term Vision

Ultimate goal: - Production-quality retro-compatible peripheral subsystem - Complete peripheral coverage for legacy platform requirements - Used in production FPGA designs - Educational resource for classic peripheral design - Foundation for mixed-vintage SoC designs

---

Version: 1.0 **Last Review:** 2025-10-29 **Next Review:** After each new block completion **Maintained By:** RTL Design Sherpa Project